

38. Face Detection using OpenCV

AIM

To detect human faces in an image using OpenCV.

CODE

```
import cv2

img = cv2.imread(r"C:\Users\chait\OneDrive\Documents\Desktop\cv lab image.jpg")

face_cascade = cv2.CascadeClassifier(
    cv2.data.harcascades + "haarcascade_frontalface_default.xml"
)

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

faces = face_cascade.detectMultiScale(gray, 1.1, 5)

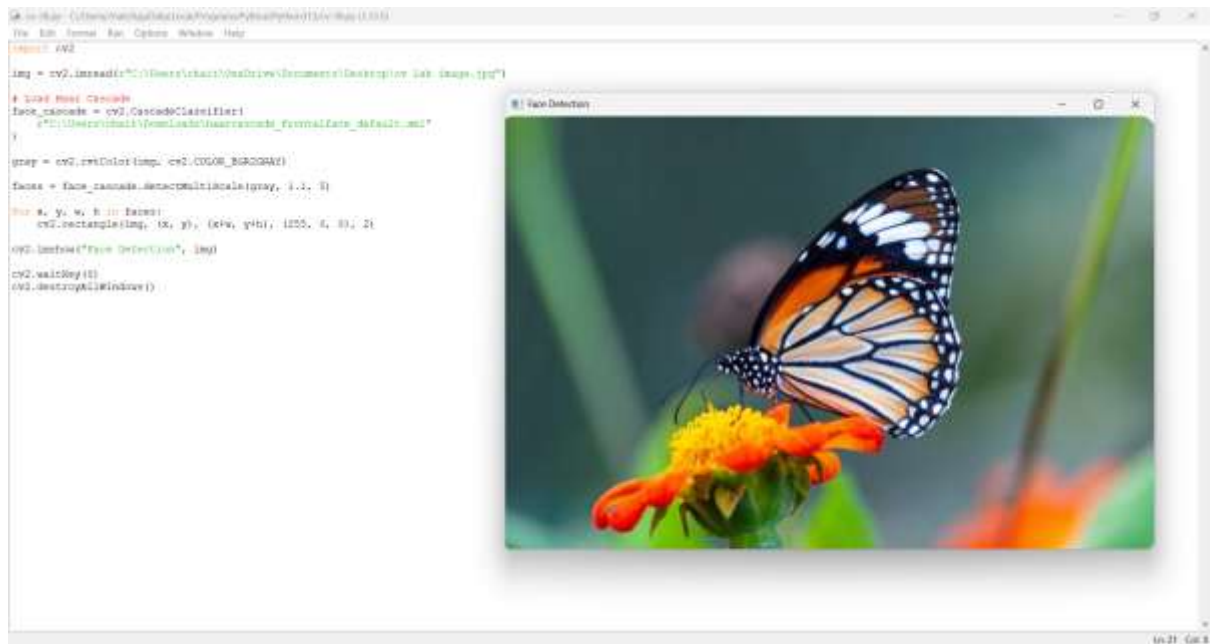
for x, y, w, h in faces:
    cv2.rectangle(img, (x, y), (x+w, y+h), (255, 0, 0), 2)

cv2.imshow("Face Detection", img)

cv2.waitKey(0)
cv2.destroyAllWindows()
```

INPUT

An image containing one or more human faces.



OUTPUT

A rectangle is drawn around the detected face(s).

RESULT

Thus, human faces were successfully detected using the Haar Cascade classifier in OpenCV.